COMPUTER ORGANIZATION

Unit-III: Instruction Codes and Control

Comprehensive Study Notes

UNIT-III: INSTRUCTION CODES AND CONTROL

1.1 What is an Instruction Code?

An **instruction code** is a group of bits that tells the computer to perform a specific operation. It consists of two parts:

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Instruction Code = OPCODE + Operand
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Part	Description	Bits
OPCODE	Specifies the operation to be performed	3-6 bits
Operand	Specifies the data or address of data	Remaining bits

1.2 Stored Program Organization

The **stored program organization** is a fundamental concept where both programs and data are stored in the same memory.

Key Features:

- Program instructions are stored in memory
- Instructions are executed sequentially
- Program counter (PC) keeps track of execution

1.3 Indirect Address

Indirect addressing means that the instruction contains the address of a memory location that holds the actual address of the data.

2.1 Common Bus Register

The **common bus register** is a shared path that connects all registers and memory units.

2.2 Basic Computer Registers

Register	Symbol	Size (bits)	Function
Program Counter	PC	12	Holds address of next instruction
Instruction Register	IR	16	Holds current instruction
Memory Address Register	MAR	12	Holds address for memory access
Memory Data Register	MDR	16	Holds data to/from memory
Accumulator	AC	16	Holds data for arithmetic/logic operations
Temporary Register	TR	16	Temporary storage
Input Register	INPR	8	Holds input data
Output Register	OUTR	8	Holds output data

3.1 Instruction Set Completeness

An instruction set is considered **complete** if it includes:

```
+-----------------------+  +-----------------------+
|   DATA TRANSFER       |  |      ARITHMETIC       |
|   Instructions        |  |   Instructions        |
+-----------------------+  +-----------------------+
+-----------------------+  +-----------------------+
|     LOGICAL           |  |       CONTROL         |
|   Instructions        |  |   Instructions        |
+-----------------------+  +-----------------------+
|   I/O Instructions    |
+-----------------------+
```

Category	Description	Examples
Data Transfer	Move data between registers/memory	LOAD, STORE
Arithmetic	Perform arithmetic operations	ADD, SUB, MUL
Logical	Perform logical operations	AND, OR, XOR
Control	Control program flow	JUMP, BRANCH, HALT
I/O	Input/Output operations	IN, OUT

3.2 Instruction Types

Type	Description	Examples
Data Transfer Instructions	Move data between registers and memory	LOAD, STORE, MOVE
Arithmetic Instructions	Perform arithmetic operations	ADD, SUB, MUL, DIV

Logical Instructions	Perform logical operations	AND, OR, XOR, NOT
Shift Instructions	Shift bits in a register	SHL, SHR
Control Instructions	Change flow of program	JMP, CALL, RET
I/O Instructions	Input and Output data	IN, OUT

3.3 Addressing Modes

Mode	Description	Example
Immediate	Operand is part of instruction	ADD #5
Direct	Instruction contains memory address of operand	LOAD 1000
Indirect	Instruction contains address of address of operand	LOAD (1000)
Register	Operand is in a register	ADD R1
Register Indirect	Register holds address of operand	ADD (R1)
Indexed	Address = base + index register	LOAD 1000(X)
Relative	Address = PC + offset	JMP +5

4.1 What is Timing and Control?

Timing and control refers to the generation of signals that coordinate and synchronize the operations of the CPU.

4.2 Control Unit Functions

Function	Description
Fetch	Gets the next instruction from memory
Decode	Determines what operation to perform
Execute	Performs the operation
Timing	Generates timing signals for synchronization
Control	Generates control signals for all components

4.3 Hardwired Control vs Microprogrammed Control

Feature	Hardwired Control	Microprogrammed Control
Implementation	Using logic gates, flip-flops, and circuits	Using microinstructions stored in control memory
Speed	Fast	Slower
Flexibility	Difficult to modify	Easy to modify
Cost	Low for simple CPUs	Higher due to control memory
Complexity	Increases with instruction set size	Easier to manage complex instruction sets

5.1 What is the Instruction Cycle?

The **instruction cycle** is the complete process of fetching, decoding, and executing an instruction.

5.2 Steps of Instruction Cycle

Step	Description	Registers Used
1. Fetch	Get instruction from memory	PC → MAR, MDR → IR
2. Decode	Interpret instruction	IR (opcode)
3. Read Operand	Get data if needed	MAR, MDR
4. Execute	Perform operation	AC, ALU
5. Store Result	Write back if needed	MDR, AC

5.3 Types of Instructions

5.3.1 Register-Reference Instructions

Register-reference instructions operate on registers rather than memory.

Instruction	Operation	Description
CLA	AC ← 0	Clear Accumulator
CMA	AC ← AC'	Complement Accumulator
CIR	AC ← Circulate Right	Rotate AC Right
CIL	AC ← Circulate Left	Rotate AC Left
INC	AC ← AC + 1	Increment Accumulator

5.3.2 Memory-Reference Instructions

Memory-reference instructions involve accessing memory.

Instruction	Operation	Description
LDA	AC ← M[address]	Load Accumulator
STA	M[address] ← AC	Store Accumulator
ADD	AC ← AC + M[address]	Add to Accumulator
SUB	AC ← AC - M[address]	Subtract from Accumulator
AND	AC ← AC & M[address]	AND with Accumulator
JMP	PC ← address	Jump to address
BRZ	IF AC=0 THEN PC ← address	Branch if Zero

5.3.3 Input/Output Instructions

Instruction	Operation	Description
IN	AC ← Input data	Read from input device
OUT	Output ← AC	Write to output device

6.1 What is Micro-Programmed Control?

Micro-programmed control is a method of implementing the control unit using microinstructions stored in a control memory.

6.2 Control Memory

Control memory stores microinstructions that define the control signals for each operation.

7.1 What is Addressing Sequencing?

Addressing sequencing is the process of determining the next microinstruction address.

7.2 Conditional Branching

Conditional branching allows the control unit to make decisions based on status conditions.

7.3 Mapping of Instructions

Mapping is the process of translating instruction codes to microinstruction addresses.

7.4 Subroutines

Subroutines are sequences of microinstructions that can be called from multiple locations.

Questions

Short Answer Questions (2-5 Marks)

1. What is an instruction code?
2. What is stored program organization?
3. What is indirect addressing?
4. What are the registers in a basic computer?
5. What is an instruction cycle?
6. What is micro-programmed control?
7. What is control memory?
8. What is conditional branching?
9. What is instruction mapping?
10. What is a subroutine?

Long Answer Questions (10 Marks)

1. Explain stored program organization with a diagram.
2. Explain instruction codes and their format.
3. Explain the registers of a basic computer.
4. Explain the instruction cycle in detail.
5. Explain micro-programmed control with a diagram.
6. Explain addressing sequencing and mapping.
7. Explain the types of instructions in a computer.
8. Explain the fetch-decode-execute cycle.

Objective Questions (1 Mark)

1. Opcode stands for:
   (a) Operation Code (b) Operational Code (c) Both (d) None
2. PC stands for:
   (a) Program Counter (b) Process Counter (c) Program Control (d) None
3. IR stands for:
   (a) Instruction Register (b) Interrupt Register (c) Both (d) None

4. The instruction cycle has:
   (a) Fetch and Execute   (b) Fetch, Decode, Execute   (c) Decode only   (d) Execute only
5. Micro-programmed control uses:
   (a) Hardwired logic   (b) Control memory   (c) Both   (d) None

QUICK REVISION

Instruction Cycle Summary

Fetch → Decode → Read Operand → Execute → Store Result

Register Summary

Register	Function
PC	Next instruction address
IR	Current instruction
MAR	Memory address
MDR	Memory data
AC	Accumulator

End of Unit-III Notes